The core

Pierre Pinson

Imperial College London, Dyson School of Design Engineering
mail: p.pinson@imperial.ac.uk - **webpage:** www.pierrepinson.com

Game Theory and Mechanism Design - Lecture 12
(This lecture heavily relies on Narahari (2014) Game Theory and Mechanism Design – chap. 28)

Core

(noun)

... the part of something that is central to its existence or character.

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- Define the **core**
- Introduce and discuss its **properties**
- **Compute** and **visualise** the core of "simple" games

Outline

1. Coalitional rationality
2. The core
3. Visualising the core
4. Computing and the core

1. **Coalitional rationality**

Individual and collective rationality

Definition (individual rationality)

In a coalitional game (N, v), a (pay-off) allocation $x = (x_1, \ldots, x_n)$ is *individually* rational if $x_i \geq v(\{i\})$, $\forall i \in N$.

Definition (collective rationality)

In a coalitional game (N, v), a (pay-off) allocation $x = (x_1, \ldots, x_n)$ is *collectively* rational if $\sum_i x_i = v(N)$.
Collective rationality yields Pareto efficiency.

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Coursework with rewards game

Consider the following characteristic function:

$v(\emptyset) = 0$

$v(\{1\}) = 50$
$v(\{2\}) = 55$
$v(\{3\}) = 65$

$v(\{1,2\}) = 120$
$v(\{2,3\}) = 136$
$v(\{1,3\}) = 144$

$v(\{1,2,3\}) = 210$

What can we say about the allocation $x = (70, 70, 70)$?

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What can we say about the allocation $x = (70, 70, 70)$?

- it is **individually** rational,
- it is **collectively** rational...
- ... but **something does not feel right**!! (to me, at least)

Coalitional properties: feasibility

Not all potential (pay-off) allocations can be attained, for all coalitions...

Definition (feasible allocation)

In a coalitional game (N, v), a (pay-off) allocation $x = (x_1, \ldots, x_n)$ is said to be *feasible* for coalition $\mathcal{C}$ if

$$\sum_{i \in \mathcal{C}} x_i \leq v(\mathcal{C})$$

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Is the allocation $x = (70, 70, 70)$ feasible...

- for $\mathcal{C} = \{1, 2, 3\}$?
- for $\mathcal{C} = \{1, 3\}$?
- for $\mathcal{C} = \{2, 3\}$?
- for $\mathcal{C} = \{1, 2\}$?

Coalitional properties: rationality

We need to generalise the concept of individual rationality.

Definition (coalitional rationality)

In a coalitional game (N, v), a (pay-off) allocation $x = (x_1, \ldots, x_n)$ is *coalitionally rational* if

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The problem with the allocation $x = (70, 70, 70)$, is that it was not coalitionally rational! (check $\{1, 3\}$)

Can you find a few allocations that are coalitionally rational?

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The problem with the allocation $x = (70, 70, 70)$, is that it was not coalitionally rational! (check $\{1, 3\}$)

Can you find a few allocations that are coalitionally rational?

- $x = (55, 65, 90)$ is coalitionally rational
- $x' = (60, 65, 85)$ is coalitionally rational
- ... actually, one could even push it, and have $x = (100, 100, 100)$ as a coalitionally rational allocation

How is this a generalisation of individual rationality?

How is this a generalisation of individual rationality?

Coalitional rationality is for all coalitions $\mathcal{C} \subseteq N$...

Though, if we restrict ourselves to coalitions of cardinal 1, i.e., $\mathcal{C} = \{i\}, \ \forall i \in N$, this defines *individual rationality*!

2. **The core**

Individually, coalitionally and collectively rational allocations

We want to find all potential allocations that could make sense for all players.

Definition (the core)

Given a coalitional game (N, v), its *core* $\mathbb{C}(N, v)$ is the set of (pay-off) allocations $x = (x_1, \ldots, x_n)$ that are both coalitionally and collectively rational, i.e.

- $\sum_{i \in \mathcal{C}} x_i \geq v(\mathcal{C}), \ \forall \mathcal{C} \subseteq N$ (coalitional rationality)
- $\sum_{i \in N} x_i = v(N)$ (collective rationality)

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Can you describe the core of that coalitional game?

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Can you describe the core of that coalitional game?

- it is a set of allocations $x = (x_1, x_2, x_3)$
- these allocations have to sum to 210, i.e., $x_1 + x_2 + x_3 = v(\{1,2,3\})$
- and we have a set of constraints (inequalities):

$$x_1 \geq v(\{1\}), \quad x_2 \geq v(\{2\}), \quad x_3 \geq v(\{3\})$$
$$x_1 + x_2 \geq v(\{1,2\}), \quad x_2 + x_3 \geq v(\{2,3\}), \quad x_1 + x_3 \geq v(\{1,3\})$$

Nash equilibrium

The core is linked to concepts we have seen before...

Proposition (the core and Nash equilibria)

Given a coalitional game (N, v) and its core $\mathbb{C}(N, v)$, any allocation $x = (x_1, \ldots, x_n)$ that is in the core, $x \in \mathbb{C}(N, v)$, is a Nash equilibrium for all players involved. They have no interest in unilaterally deviating from it.

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Let us focus on the allocation $x = (55, 65, 90)$... how is that a Nash equilibrium?

- if player 1 decreases their allocation (e.g., 54), they would be worse off. And, if they increase it (e.g., 56), it is not feasible anymore, and everybody is worse off...

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- if player 1 decreases their allocation (e.g., 54), they would be worse off. And, if they increase it (e.g., 56), it is not feasible anymore, and everybody is worse off...
- if player 2 decreases their allocation (e.g., 64), they would be worse off. And, if they increase it (e.g., 66), it is not feasible anymore, and everybody is worse off...

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Given a coalitional game (N, v) and its core $\mathbb{C}(N, v)$, any allocation $x = (x_1, \ldots, x_n)$ that is in the core, $x \in \mathbb{C}(N, v)$, is a Nash equilibrium for all players involved. They have no interest in unilaterally deviating from it.

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- if player 1 decreases their allocation (e.g., 54), they would be worse off. And, if they increase it (e.g., 56), it is not feasible anymore, and everybody is worse off...
- if player 2 decreases their allocation (e.g., 64), they would be worse off. And, if they increase it (e.g., 66), it is not feasible anymore, and everybody is worse off...
- if player 3 decreases their allocation (e.g., 89), they would be worse off. And, if they increase it (e.g., 91), it is not feasible anymore, and everybody is worse off...

Properties of the core

It cannot just be anything...

Proposition (convexity and compactness)

Given a coalitional game (N, v) with non-empty core $\mathbb{C}(N, v)$, the core is convex and compact, i.e.

- $x, x' \in \mathbb{C}(N, v) \Longrightarrow \alpha x + (1 - \alpha)x' \in \mathbb{C}(N, v)$, $\forall \alpha \in [0, 1]$ *(convexity)*
- $\exists M : x_i \leq M$, $\forall i$ *(compactness)*

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Let us illustrate these 2 properties:

- Compactness

  If I define M as $M = v(\{1,2,3\})$, then for all potential allocations we would have $x_i \leq M, \ i = 1, 2, 3$.

- Convexity

  We know of 2 allocations that are in the core, i.e. $x = (55, 65, 90)$ and $x' = (60, 65, 85)$.
  If I define a new allocation as $\frac{1}{2}(x + x') = (57.5, 65, 87.5)$, it is also in the core.

Emptiness

Maybe there is simply no single allocation that makes sense for all players

Proposition (Empty core)

Considering all potential coalitional games (N, v), some of them have a non-empty core $\mathbb{C}(N, v) \neq \emptyset$, but some of them have an empty core, i.e., $\mathbb{C}(N, v) = \emptyset$

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Coursework with rewards game (but with a twist)

Consider the following characteristic function:

$v(\emptyset) = 0$	$v(\{1\}) = 50$	$v(\{1, 2\}) = 120$	$v(\{1, 2, 3\}) = 180$
	$v(\{2\}) = 55$	$v(\{2, 3\}) = 136$	
	$v(\{3\}) = 65$	$v(\{1, 3\}) = 144$	

Its core is empty! (there is no allocation that can be both coalitionally and collectively rational)

3. **Visualising the core**

Proposition (normalisation)

Given a coalitional game (N, v), it is always possible to define a normalised version (N, w) of the game, with

$$w(\mathcal{C}) = \frac{1}{v(N) - \sum_{i \in N} v(\{i\})} \left(v(C) - \sum_{i \in \mathcal{C}} v(\{i\}) \right), \ \forall \mathcal{C} \subseteq N$$

The resulting normalised game (N, w) is such that

- $w(\{i\}) = 0, \ \forall i \in N$
- $w(N) = 1$

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Coursework with rewards game (to be normalised)

$$w(\emptyset) = 0 \qquad w(\{1\}) = 0 \qquad w(\{2\}) = 0 \qquad w(\{3\}) = 0$$

$$w(\{1,2\}) = \frac{1}{40}(v(\{1,2\}) - v(\{1\}) - v(\{2\})) = \frac{3}{8}$$

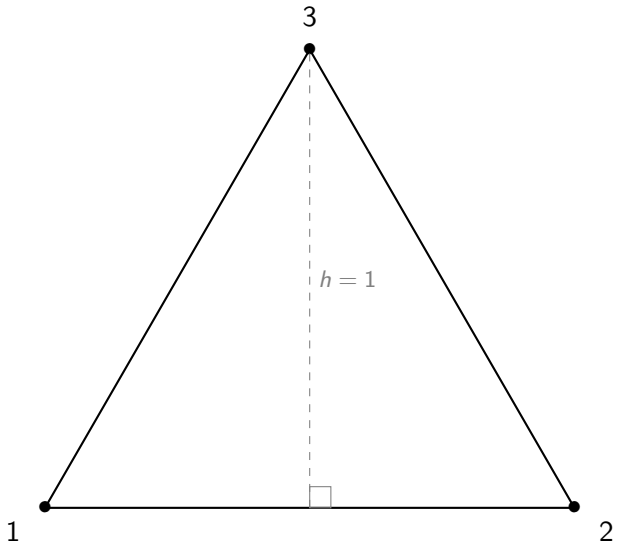
$$w(\{2,3\}) = \frac{1}{40}(v(\{2,3\}) - v(\{2\}) - v(\{3\})) = \frac{2}{5}$$

$$w(\{1,3\}) = \frac{1}{40}(v(\{1,3\}) - v(\{1\}) - v(\{3\})) = \frac{29}{40}$$

$$w(\{1,2,3\}) = 1$$

Graphical representation

The reason why we want to normalise games is that we can them look at them based on an *equilateral triangle* of height 1... (in the case of 3 players)



Why is that a good idea to use such equilateral triangle as a basis?

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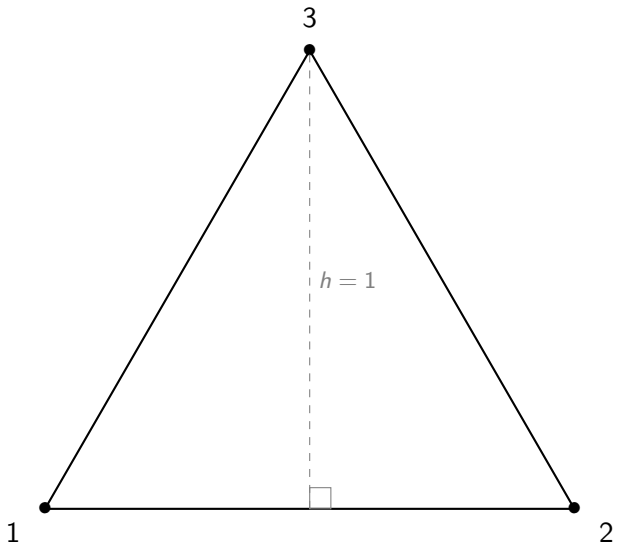

Why is that a good idea to use such equilateral triangle as a basis?

For instance, look at the 3 points defining the triangle:

- one of them defines the allocation (1,0,0)
- one of them defines the allocation (0,1,0)
- one of them defines the allocation (0,0,1)

And, these are imputations
(i.e., individually and collectively rational)

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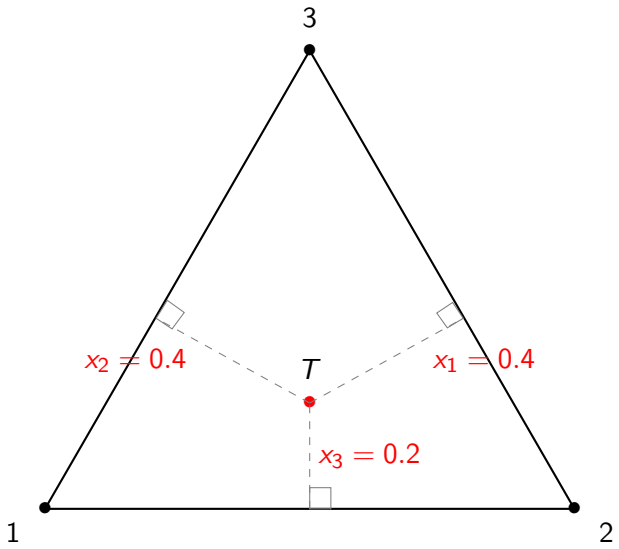
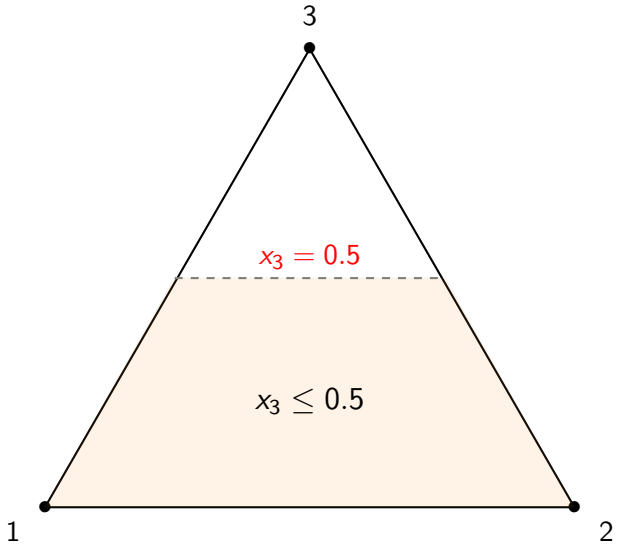


All the points inside the triangle are imputations!

(as for the reduced feasibility set in Nash bargaining)

Adding constraints within the triangle

The triangle describe the set of imputations... The core is a smaller set of collectively rational imputations.



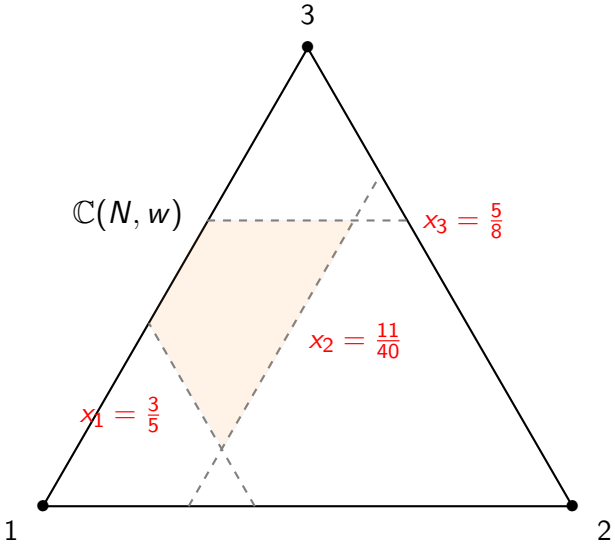
Constraints can be added within the triangle

For instance,

- if we say that $x_1 + x_2 \geq 0.5$
- it means that $x_3 \leq 0.5$
- we can add the line to the triangle
- and obtain the resulting area that respects that constraint (in orange, here)

Representing the core of the coursework with rewards game

We use the normalised version of the game as a basis



- $x_i \geq w(\{i\})$, $\forall i$, is satisfied for all points (individual rationality)
- $\sum_i x_i = 1$ is satisfied for all points (collective rationality)
- we add the coalitional rationality constraints, i.e.

$$x_1 + x_2 \geq \frac{3}{8} \Longrightarrow x_3 \leq \frac{5}{8}$$

$$x_2 + x_3 \geq \frac{2}{5} \Longrightarrow x_1 \leq \frac{3}{5}$$

$$x_1 + x_3 \geq \frac{29}{40} \Longrightarrow x_2 \leq \frac{11}{40}$$

- the resulting area is the core of the game!

4. **Computing and the core**

Characterising the core

We can use a linear program to characterise the core

Definition (Coalitionally rational LP)

Given a coalitional game (N, v), for the set of players $N = \{1, \ldots, n\}$, the following *linear program* (LP) allows telling whether the core is empty or not:

$$
\begin{aligned}
x^* = \underset{x}{\operatorname{argmin}} \quad & x_1 + \ldots + x_n \\
\text{s.t.} \quad & \sum_{i \in \mathcal{C}} x_i \geq v(\mathcal{C}), \ \forall \mathcal{C} \subseteq N \\
& x_1, \ldots, x_n \geq 0
\end{aligned}
$$

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& x_1, \ldots, x_n \geq 0
\end{aligned}
$$

Coursework with rewards game

This linear program would be written as:

$$
\begin{aligned}
x^* = \underset{x}{\operatorname{argmin}} \quad & x_1 + \ldots + x_n \\
\text{s.t.} \quad & x_1 \geq 50, \ x_2 \geq 55, \ x_3 \geq 65 \\
& x_1 + x_2 \geq 120 \\
& x_2 + x_3 \geq 136 \\
& x_1 + x_3 \geq 144
\end{aligned}
$$

How do we use this linear program?

Firstly, one should noticed that there should always be at least one solution x^* to this linear program if it is feasible (i.e., the region defined by the constraints is not empty)

$$\sum_{i \in N} x_i^* \quad > \quad v(N)$$

There is a solution, but it is not collectively rational... Hence, the core is empty

$$\sum_{i \in N} x_i^* \quad = \quad v(N)$$

There is one or more solutions, which are both coalitionally and collectively rational... the set of solutions define the core!

Wrapping up

- The core is the set of **Nash equilibria** of coalition games, which are both coalitionally and collectively rational
- Coalitional games can be **normalised**
- One can always **visualise** a 3-person coalitional game
- The core can be **empty**
- **Optimisation** can be used to characterise the core

Further readings (/lecture notes for this topic):

Narahari Y (2014) *Game Theory and Mechanism Design* - chap. 28 (available on Blackboard Learn)

Thanks for your attention!